

Improving English-to-Indic Translation for Very Low-Resource Languages using SLMs: A Study on Devanagari-Script Languages Beyond Hindi

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Abstract

This work examines the use of small and mid-sized open-source language models (under 10B parameters) for English-to-Indic Machine Translation in low-resource settings. The study focuses on five underrepresented Devanagari-script languages: Awadhi, Bhojpuri, Chhattisgarhi, Magahi, and Maithili, which are often grouped with Hindi despite having distinct linguistic features and cultural identities. We compare the performance of multiple small language models and further fine-tune one representative model using instruction-style prompting with parameter-efficient fine-tuning techniques. Translation quality is evaluated on the FLORES-200 benchmark using BLEU and ChrF metrics. The findings indicate that fine-tuned models significantly outperform baseline models, alongside evident variations across different models. Overall, the study suggests that smaller open-source models can provide an efficient and practical solution for developing translation systems for underrepresented Indic languages.

Keywords: Machine Translation, Small Language Models, Low-Resource Languages, Indic Languages, Devanagari Script, Parameter-Efficient Fine-Tuning

1 Introduction

Large Language Models (LLMs) have significantly reshaped Natural Language Processing (NLP), enabling systems to not only generate text but also reason and operate across multiple languages with a level of fluency that was previously difficult to achieve. Models such as GPT (Brown et al. 2020), LLaMA (Grattafiori et al. 2024), and Gemma (Gemma Team et al. 2025) perform strongly across a variety of generation tasks, including summarisation, question answering, and machine translation (MT). However, these capabilities are not evenly distributed across languages. In practice, performance remains heavily concentrated in a small set of high-resource languages, primarily because most pretraining data is in English. As a result, while these models produce highly fluent English outputs, their representations of low-resource languages tend to remain shallow, making accurate translation into such languages a persistent challenge.

Recent work has begun exploring Small Language Models (SLMs), typically defined as open-source models with fewer than ten billion parameters, as a more practical and resource-efficient alternative to large, closed systems (Subramanian et al. 2025; Belcak et al. 2025). Few model variants of Llama, Gemma and Qwen are particularly notable at this scale: they offer strong instruction-following abilities while remaining feasible to fine-tune on modest hardware (Grattafiori et al. 2024; Gemma Team et al. 2025). That said, their multilingual performance still reflects the same imbalance seen in larger models. In particular, their behaviour on extremely low-resource languages is not yet well understood.

This limitation becomes visible in the Indian context because of its rich linguistic diversity. The Devanagari script is shared by several Indo-Aryan languages. In the field of NLP, Hindi has received the most attention, whereas related languages such as Awadhi, Bhojpuri, Chhattisgarhi, Magahi, and Maithili remain significantly underrepresented (Mundotiya et al. 2021; Kumar et al. 2022). These languages are often grouped under Hindi in practice, but they differ in meaningful linguistic and cultural ways, making such grouping problematic for MT research. This leads to an interesting imbalance: despite having large speaker bases, these languages have very limited digital presence, and consequently receive little attention in modern language technologies.

Another important consideration is the asymmetry in real-world translation needs. In many practical settings including education, governance, and local content creation, the primary requirement is to translate content from English into regional languages ($EN \rightarrow xx$), rather than the reverse. This is also aligned with how current models are trained: since they are predominantly exposed to English during pretraining, they tend to have stronger comprehension on the source side. Translating from English therefore leverages this strength, while generation in the target language can still benefit from cross-lingual transfer, particularly due to shared scripts and lexical overlap with Hindi. For these reasons, this work focuses exclusively on the $EN \rightarrow xx$ translation direction.

Despite growing interest in LLM-based MT and parameter-efficient fine-tuning for low-resource languages (reviewed in Section 2), there is still no systematic study

focusing on languages such as Awadhi, Bhojpuri, Magahi, and Maithili using modern open-source SLMs.

In this work, we study five low-resource Devanagari-script languages and benchmark several open-source SLMs: *Gemma 3*, *Gemma 4*, *LLaMA 3.1*, *LLaMA 3.2*, *Qwen 3.5*, *TranslateGemma*, and *SarvamTranslate* against encoder-decoder neural machine translation models, *IndicTrans2* and *NLLB 200*. We evaluate parameter-efficient adaptation under extreme low-resource conditions (fewer than 1,000 sentence pairs per language), and analyze how fine-tuning affects translation behaviour across closely related dialects. In addition to quantitative metrics, we report findings from linguistic analysis of model outputs. Our results show that even with limited data and compute, fine-tuned SLMs can achieve substantial improvements in translation quality.

The main contributions of this paper are as follows:

- A systematic benchmark of seven open-source SLMs ($< 10\text{B}$ parameters) against dedicated NMT systems across five underexplored Devanagari-script languages in the $\text{EN} \rightarrow \text{xx}$ setting.
- Empirical evidence that parameter-efficient fine-tuning with fewer than 1,000 sentence pairs per language yields consistent gains, substantially narrowing the gap with specialised translation systems.
- A linguistic analysis of model outputs showing how fine-tuning improves language use, dialect consistency, and reduces cross-dialect mixing beyond standard metrics.
- A study of adaptation strategies, including multilingual versus monolingual fine-tuning and LoRA configurations, providing practical guidance for low-resource MT.
- Release of fine-tuned model checkpoints to support reproducibility and further research on underrepresented Indic languages.

2 Related Work

2.1 Large Language Models for Machine Translation

The advent of large scale pre-trained language models has fundamentally shifted the paradigm in machine translation. Models such as GPT (Brown et al. 2020), LLaMA (Grattafiori et al. 2024), and Gemma (Gemma Team et al. 2025) have demonstrated strong cross lingual capabilities along with in-context learning emerging as a practical approach for MT without task-specific training (Vilar et al. 2023; Agrawal et al. 2023; Bawden and Yvon 2023). Hendy et al. (2023) conducted a comprehensive evaluation of GPT models on MT, finding performance comparable to dedicated translation systems for high resource language pairs. However, these impressive zero-shot capabilities do not generalise well to low-resource settings, where the lack of pre-training data creates a significant performance gap. Singh et al. (2024) benchmarked LLMs including PaLM-2 (Anil et al. 2023), GPT-4, and LLaMA on Indic language generation tasks, finding that even the strongest models struggle on low-resource language variants.

2.2 Small Language Models

The growing emphasis on computational efficiency and on device deployment has spurred significant interest in Small Language Models (SLMs), broadly defined as open models with fewer than ten billion parameters (Subramanian et al. 2025; Belcak et al. 2025). SLMs are five to ten times smaller than their large counterparts and are optimised for high volume processing on resource constrained devices (Abdin et al. 2024). SLMs can be fine-tuned to achieve high accuracy for tasks in a data-constrained domain, with fewer GPUs and less time than comparable large model fine-tuning.

Within the SLM landscape, the models evaluated in this study span a diverse set of architectures and training objectives. Gemma 3 and 4 were specifically designed to run on standard consumer grade hardware, with a post training recipe that substantially improves multilingual abilities, making Gemma 3 4B competitive with the much larger Gemma 2 27B across benchmarks (Gemma Team et al. 2025). Qwen3.5 (Yang et al. 2025) and LLaMA 3.2 3B represent further points on the parameter efficiency frontier, while TranslateGemma 4B and SarvamTranslate 4B are specialised models fine-tuned for machine translation (Gala et al. 2023). Together, these models allow a systematic analysis of how model size, pre-training data scale, and task specialization impact in very low-resource settings.

Comparative benchmarking of open source SLMs underlines the appeal of this model tier for resource constrained research. A systematic review of SLMs published between 2023 and 2025 confirms LLaMA and Gemma derivatives as the dominant paradigm for models up to 7 billion parameters, highlighting their widespread adoption across compression and optimization benchmarks (Subramanian et al. 2025). Despite these advances, the multilingual capabilities of SLMs in the sub-10B range remain heavily skewed toward European and East Asian languages; their performance on morphologically rich, extremely low-resource languages is yet to be thoroughly characterised.

2.3 Fine-Tuning LLMs for Translation

While prompt-based approaches are convenient and ready to use, fine-tuning remains the most reliable method for specialising LLMs on downstream tasks. Full fine-tuning of billion-parameter models is often computationally prohibitive in low-resource settings, making Parameter-Efficient Fine-Tuning (PEFT) (Houlsby et al. 2019) techniques increasingly attractive. LoRA (Hu et al. 2021), which injects trainable low rank matrices into frozen transformer layers, has emerged as the dominant PEFT approach, drastically reducing trainable parameters without sacrificing performance. Iyer et al. (2023) showed that fine-tuning improves LLM performance on ambiguous translation cases. Xu et al. (2023) adapted LLaMA for translation via continued monolingual pre-training followed by parallel data fine-tuning across German, Czech, Russian, Chinese, and Icelandic. Alves et al. (2024) extended this to a multilingual setting across nine European languages using LLaMA-2. While these works demonstrate the effectiveness of fine-tuning, they focus exclusively on relatively high resource languages. Our work extends this line of research to a truly low-resource scenario, applying LoRA fine-tuning using the FLORES-200 dev set (NLLB Team

et al. 2022) with only 997 sentences per language to test the sample efficiency limits of PEFT for very low-resource Indic MT.

2.4 Evaluation Metrics for Low-Resource Machine Translation

Automatic evaluation in low-resource MT is particularly sensitive to metric choice. BLEU (Papineni et al. 2002) evaluates word-level n-gram precision with a brevity penalty and remains widely used for its interpretability and comparability across studies. In contrast, ChrF (Popović 2015) measures character-level n-gram F-score, making it more robust to morphological variation, inflectional diversity, and diacritic (matra) variation common in Devanagari-script languages.

These metrics exhibit complementary behavior in extremely low-resource settings. Prior work on Indic languages such as Magahi, Bhojpuri, and Chhattisgarhi (Kumar et al. 2026) shows that ChrF often yields higher absolute scores due to surface-form overlap, while BLEU provides stronger signals of lexical precision.

Neural metrics such as COMET (Rei et al. 2022) depend on pretrained multilingual encoders such as XLM-R (Conneau and Lample 2019), whose coverage of several low-resource Indic languages remains limited, potentially affecting their reliability.

2.5 Machine Translation for Indic Languages

Indic language MT has seen significant progress through large scale supervised NMT systems. The NLLB project (NLLB Team et al. 2022) introduced a multilingual translation model covering 200 languages with dedicated coverage of low-resource Indic languages including all five languages being investigated in this work. Gala et al. (2023) released IndicTrans2, a sub-2B model, achieving high quality MT across all 22 official languages of India. At the corpus level, the Bharat Parallel Corpus Collection (BPCC) (Gala et al. 2023) provided 230 million sentence pairs spanning 22 languages, including several Devanagari script dialects. Recent efforts have also focused on developing dedicated parallel corpora for specific low-resource language pairs, for instance, MaithiliMT introduces a multi domain Hindi-Maithili parallel corpus to support machine translation research for this underrepresented language pair (Appicharla et al. 2026). For the specific languages studied here, Mundotiya et al. (2021) constructed the first large annotated corpora for Bhojpuri, Magahi, and Maithili, while Kumar et al. (2022) introduced a transfer-learning-based semi-supervised pseudo-corpus generation approach that improved BLEU by up to +15.56 points for Bhojpuri-Hindi zero-shot translation.

Despite this progress, the application of LLMs to this long tail of Devanagari script languages remains largely unexplored. Dash and Sharma (2025) benchmarked Gemma 1.1, GPT-3.5, and LLaMA-3 on English(EN) $\leftrightarrow$ {Hindi, Bengali, Odia, Chhattisgarhi}(xx) translation and fine-tuned LLaMA-3 8B with LoRA, reporting consistent gains across all eight directions. Their evaluation found that $xx \rightarrow EN$ scores were systematically higher than $EN \rightarrow xx$, attributing this asymmetry to the dominance of English in LLM pretraining, and that Gemma 1.1 performed particularly poorly on Odia while GPT-3.5 was the strongest zero-shot baseline on Hindi and Chhattisgarhi. Gupta (2025) confirmed in a comprehensive survey of South Asian NLP (2022-2024)

that NMT translation for Hindi dialects is nearly absent in the LLM literature. To the best of our knowledge, systematic benchmarking and PEFT-based approaches for improving translation in Maithili, Magahi, Chhattisgarhi, Bhojpuri, and Awadhi with modern open-source SLMs remain underexplored, a gap this study directly addresses.

3 Experimental Setup

3.1 Target Languages: Underrepresented Devanagari-Script Languages

The five languages studied in this work, Awadhi, Bhojpuri, Chhattisgarhi, Magahi, and Maithili belong to the Indo-Aryan branch of the Indo-European language family and are spoken primarily across the northern and central regions of the Indian sub-continent. All five languages use the Devanagari script, which they share with Hindi, and this surface-level similarity has historically led to their misclassification as Hindi in both popular usage and NLP research. However, each language exhibits distinct phonological, morphological, and lexical features that justify independent treatment for translation purposes (Mundotiya et al. 2021; Kumar et al. 2022).

Awadhi is spoken by approximately 3-4 million people (Eberhard et al. 2024), concentrated in the Awadh region of the Indian state of Uttar Pradesh. It is historically notable as the language of the medieval literary classic *Ramcharitmanas*, and its verb morphology differs in meaningful ways from Standard Hindi.

Bhojpuri has one of the largest speaker bases among the five, with estimates ranging from 50 to 60 million speakers across the Indian states of Uttar Pradesh and Bihar, and the diaspora (Eberhard et al. 2024). Despite its wide spoken use, it remains severely underrepresented in digital text corpora. Its case system and aspectual verb constructions diverge substantially from Hindi (Mundotiya et al. 2021).

Chhattisgarhi is the primary language of the Indian state of Chhattisgarh and is spoken by approximately 13 million people (Eberhard et al. 2024). It exhibits postpositional case marking and a phonological inventory, including retroflex sounds, that is absent in Standard Hindi. This makes straightforward Hindi-to-Chhattisgarhi transfer unreliable for NMT systems (Dash and Sharma 2025).

Magahi is spoken by around 13-18 million people (Eberhard et al. 2024), predominantly in the Indian states of Bihar and Jharkhand. It is characterised by an elaborate honorific system and morphological distinctions in verb agreement that do not map directly onto Hindi grammatical categories (Mundotiya et al. 2021).

Maithili has the largest literary tradition among the five, with written records dating to the 14th century. It is spoken by approximately 34 million people (Eberhard et al. 2024) in Bihar state of India and the Terai region of Nepal and holds the status of an official language in India. Maithili’s distinct script history and complex pronominal system differentiate it clearly from Hindi (Appicharla et al. 2026).

Despite these distinct identities and sizeable speaker populations, all five languages have minimal digital presence and very limited parallel corpora. Their absence from mainstream pretraining datasets explains the low coverage they receive from general-purpose LLMs, making them a representative and challenging test bed for extremely low-resource MT research.

3.2 Dataset and Benchmark

Evaluation benchmark

We use the FLORES-200 benchmark (NLLB Team et al. 2022) as our evaluation dataset. FLORES-200 is a professionally translated, sentence-aligned, multi-domain evaluation corpus covering 200 languages. It was constructed from English Wikipedia and Wikinews passages and translated by professional translators, ensuring consistent quality across all language pairs. The test set for each language contains 1,012 sentences spanning a variety of domains including science, travel, and everyday topics. For all five target languages considered here, Awadhi, Bhojpuri, Chhattisgarhi, Magahi, and Maithili, we evaluate on the full FLORES-200 test split, which has become the de facto standard for low-resource MT evaluation (Gala et al. 2023; Dash and Sharma 2025).

Fine-tuning data

To simulate an extreme low-resource setting, we restrict fine-tuning to the FLORES-200 development set (997 sentences per language). No additional parallel corpora, such as Samanantar (Ramesh et al. 2022) or BPCC, are used. This constraint is intentional, as it probes the sample efficiency of parameter-efficient fine-tuning under conditions that closely reflect realistic data availability for these languages. Evaluation is conducted on the FLORES-200 *test set*, which remains disjoint from the fine-tuning data.

Evaluation metrics

We evaluate translation quality using BLEU and ChrF. BLEU (Papineni et al. 2002) measures modified precision over word-level n-grams (up to 4) with a brevity penalty and is computed using the *sacrebleu* Python library for reproducibility. ChrF (Popović 2015) computes character-level n-gram F-score and is less sensitive to morphological variation and tokenisation differences.

We report both metrics to provide complementary perspectives and to enable comparison with prior work. We do not report COMET (Rei et al. 2022) due to limited support of current pretrained encoders for the low-resource Indic languages understudy.

4 Methodology

This section outlines our experimental setup, including the selected pre-trained small language models (SLMs), the prompting and fine-tuning procedure, and key implementation details.

4.1 Pre-trained SLMs for Baseline Translation

We evaluate a set of small language models (SLMs, <10B parameters) alongside dedicated neural machine translation (NMT) systems. Table 1 summarises the models, including their scale and whether they are general-purpose or translation-focused.

Table 1: Models evaluated for baseline translation.

Model	Params	Type	Category
Gemma 3	4B	General-purpose	LLM
Gemma 4	8B	General-purpose	LLM
LLaMA 3.1	8B	General-purpose	LLM
LLaMA 3.2	3B	General-purpose	LLM
Qwen 3.5	9B	General-purpose	LLM
TranslateGemma	4B	Translation-focused	LLM (fine-tuned)
SarvamTranslate	4B	Translation-focused	LLM (fine-tuned)
IndicTrans2	1B	Translation-focused	NMT
NLLB-3.3B	3.3B	Translation-focused	NMT

4.2 Instruction-Based Translation Prompting

For general-purpose decoder-only LLMs, translations are generated using explicit natural-language instructions in a chat-style two-message format. In the **zero-shot** setting, each model receives a system message that defines the task and language pair, followed by a user message containing the source sentence:

```
messages = [
  {
    "role": "system",
    "content": "Don't provide any justification or extra "
              "output, just the translated text. Translate "
              "the following sentence from {src_lang} to "
              "{tgt_lang}:"
  },
  {
    "role": "user",
    "content": "{source sentence}"
  }
]
```

In the message template, `src_lang` is fixed to English and `tgt_lang` ranges over the five Devanagari-script target languages. The system message constrains the model to output only the translation, reducing extraneous content such as explanations or multiple candidates.

Specialised MT systems such as NLLB and IndicTrans2 do not use instruction prompting. Instead, translation is performed via standard pipelines using source and target language codes. This distinction is important for interpreting differences in baseline performance across model types.

4.3 Fine-Tuning with Instruction Prompts

Based on baseline evaluation results, we select **Gemma 4 E4B** for parameter-efficient fine-tuning, balancing translation performance and feasibility on a single GPU.

Table 2: Chat-style instruction-following supervision used for fine-tuning. Each instance pairs a *system* role, a *user* directive containing the English source, and an *assistant* turn with the target translation. The source language is fixed to English and [Language] ranges over the five Devanagari-script targets; a Maithili example is shown.

Role	Content
System	You are a helpful and literal translator. Translate faithfully without adding extra words.
User	Translate from English to [Language] (here: Maithili). Source: Boomerang throwing is a popular skill that many tourists want to acquire. Answer with only the translation.
Assistant	बुमेरांग फेंकनाई एकटा लोकप्रिय कौशल अछि जाहि केँ कतेक पर्यटक सिखय चाहैत छैथ।

Training Data and Instruction Format

Fine-tuning uses the FLORES-200 dev split (NLLB Team et al. 2022) (997 sentences per language), simulating an extreme low-resource setting. Data from all five language pairs is combined into a multilingual training set of 4,985 sentence pairs. Each instance is formatted as a chat-style instruction: a *system* message defines the translator role, a *user* message provides the translation directive and English source, and the *assistant* turn contains the target reference (Table 2).

During fine-tuning, a fixed instruction template is used to structure training examples, while inference for both baseline and fine-tuned models is conducted using a separate prompt template.

LoRA Fine-Tuning

We apply **Low-Rank Adaptation (LoRA)** (Hu et al. 2021), a parameter-efficient fine-tuning (PEFT) method that freezes pretrained weights and introduces trainable low-rank matrices into Transformer layers. For a weight matrix $W_0 \in \mathbb{R}^{d \times k}$, updates are parameterised as:

$$W = W_0 + \frac{\alpha}{r} \mathbf{B} \mathbf{A}, \quad \mathbf{B} \in \mathbb{R}^{d \times r}, \quad \mathbf{A} \in \mathbb{R}^{r \times k}$$

Here, r denotes the rank of the low-rank decomposition, controlling the number of additional trainable parameters, while the scaling factor α determines the magnitude of the LoRA update. We additionally evaluate multiple configurations of r and α as part of the ablation study (Section 5.3).

This formulation reduces the number of trainable parameters, enabling efficient fine-tuning while preserving pretrained representations.

Multilingual Fine-Tuning

We adopt a **multilingual many-to-many** strategy by combining all five directions (EN→Awadhi, Bhojpuri, Chhattisgarhi, Magahi, Maithili) into a single shuffled dataset. This promotes shared cross-lingual representations and improves generalisation under limited data.

We also conduct a monolingual training ablation (Section 5.3) to isolate the effect of multilingual training.

4.4 Implementation Details

All experiments are conducted on a **single NVIDIA A100 40 GB GPU** using the HuggingFace **Transformers** and **PEFT** libraries. All experiments are run with fixed random seeds to ensure reproducibility.

As all target languages use the Devanagari script, no additional preprocessing is required beyond standard tokenisation.

Training hyperparameters used for LoRA fine-tuning are summarised in Table 3.

Table 3: Hyperparameters used for LoRA fine-tuning of Gemma 4 E4B.

Parameter	Value
Learning rate	5×10^{-5}
Max seq length	512
Train batch size	2
Gradient accumulation steps	4
Total effective batch size	8
Optimiser	AdamW
LR scheduler	Cosine
Warmup ratio	0.05
Weight decay	0.01
Number of training epochs	3
LoRA rank (r)	16
LoRA scaling (α)	32
Hardware	1× NVIDIA A100 40 GB

5 Results and Analysis

5.1 Baseline Translation Performance of SLMs

This section presents baseline translation performance across models and languages. We primarily analyse results using ChrF (Table 4), as it better captures lexical and morphological variation in low-resource settings, and use BLEU (Table 5) as complementary evidence. For clarity, models are grouped into general-purpose SLMs and task-specific translation systems.

Table 4: Baseline translation performance (EN $\rightarrow$ xx) on FLORES-200 (ChrF). **Bold:** best SLM; underline: overall best; (-) unsupported pairs.

Language	General-Purpose SLMs					Task-Specific Models			
	Gemma3 4B	Gemma4 E4B	Llama3.1 8B	Llama3.2 3B	Qwen3.5 9B	Sarvam Translate 4B	Translate Gemma 4B	IT2 1B	NLLB-200 3.3B
Awadhi	39.52	35.20	5.74	14.40	35.63	-	41.23	-	<u>43.74</u>
Bhojpuri	30.93	35.82	4.42	19.53	27.59	-	34.65	-	<u>43.25</u>
Chhattisgarhi	33.59	37.23	38.03	1.63	36.05	-	41.75	-	<u>54.74</u>
Magahi	41.67	37.01	38.94	13.06	38.99	-	45.41	-	<u>57.69</u>
Maithili	27.10	34.29	30.71	23.93	35.65	51.24	37.37	<u>54.52</u>	47.77
Average	34.56	35.91	23.57	14.51	34.78	-	40.08	-	<u>49.44</u>

Across all languages, task-specific models consistently outperform general-purpose SLMs. In particular, NLLB-200 achieves the strongest overall performance, obtaining the highest average ChrF score (49.44) and leading in four of the five language pairs. This trend is mirrored in BLEU, where NLLB-200 again achieves the highest average (26.40).

Within the general-purpose SLM group, Gemma variants perform most competitively. Gemma-4 (E4B) achieves the best average performance among SLMs (35.91 ChrF), improving over Gemma-3 (4B), suggesting benefits from scale and architectural refinements. Qwen3.5 remains competitive but does not surpass Gemma-4 on any language, while the LLaMA family exhibits weaker and less stable performance.

Performance also varies substantially across languages. Magahi and Chhattisgarhi generally yield higher scores across both model groups, whereas Bhojpuri remains more challenging, with consistently lower BLEU and ChrF scores. Maithili presents a notable exception: while general-purpose SLMs achieve moderate performance, smaller task-specific models such as IndicTrans2-1B and SarvamTranslate-4B achieve very high ChrF scores (54.52 and 51.24, respectively), outperforming even larger models. This suggests that targeted training data and modelling choices can outweigh model scale for certain languages.

Overall, the results demonstrate a clear performance gap between general-purpose SLMs and specialised MT models, establishing substantial headroom for improvement through fine-tuning, as explored in subsequent sections.

5.2 Fine-Tuned Translation Performance of Gemma-4

This section presents translation performance after supervised fine-tuning, summarised in Table 6 and complemented by the aggregate comparison in Figure 1. To assess the robustness of the observed gains, we performed paired bootstrap resampling, comparing the fine-tuned model against the Gemma 4 E4B baseline.

Table 5: Baseline translation performance (EN $\rightarrow$ xx) on FLORES-200 (BLEU). **Bold:** best SLM; underline: overall best; (-) unsupported pairs.

Language	General-Purpose SLMs					Task-Specific Models			
	Gemma3 4B	Gemma4 E4B	Llama3.1 8B	Llama3.2 3B	Qwen3.5 9B	Sarvam Translate 4B	Translate Gemma 4B	IT2 1B	NLLB-200 3.3B
Awadhi	15.38	14.70	2.16	4.17	13.66	-	16.16	-	<u>21.81</u>
Bhojpuri	11.09	15.08	1.54	5.72	9.72	-	12.46	-	<u>21.54</u>
Chhattisgarhi	11.09	15.36	14.27	0.71	13.04	-	16.28	-	<u>30.20</u>
Magahi	17.63	16.30	16.14	4.44	14.76	-	20.49	-	<u>35.26</u>
Maithili	6.48	12.38	7.25	6.86	11.00	26.08	12.20	<u>29.49</u>	23.20
Average	12.33	14.76	8.27	4.38	12.43	-	15.52	-	<u>26.40</u>

Table 6: Translation performance of fine-tuned Gemma 4 E4B on low-resource Indic languages (FLORES-200, EN $\rightarrow$ xx).

Language Pair	ChrF	BLEU
English $\rightarrow$ Awadhi	45.60	22.69
English $\rightarrow$ Bhojpuri	37.60	17.29
English $\rightarrow$ Chhattisgarhi	44.58	21.19
English $\rightarrow$ Magahi	45.50	22.90
English $\rightarrow$ Maithili	40.56	16.21
Average	42.77	20.06

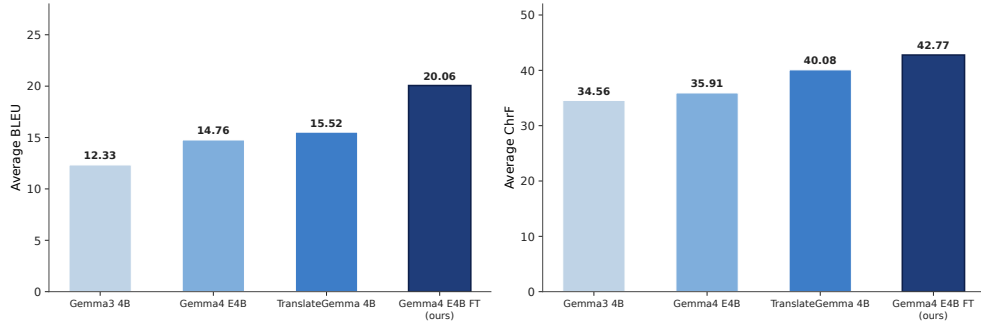


Figure 1: Average BLEU (left) and ChrF (right) for Gemma-family models on five Devanagari languages (EN $\rightarrow$ xx, FLORES-200), with fine-tuned Gemma 4 E4B (ours) performing best.

Fine-tuning leads to substantial performance improvements over the zero-shot baseline for general-purpose SLMs. In particular, the fine-tuned Gemma-4 (E4B) model achieves an average ChrF score of 42.77 and an average BLEU score of 20.06, representing significant gains of +19% and +36% in ChrF and BLEU respectively over the baseline Gemma-4 (E4B) model. Within the Gemma family, compared to

Gemma-3 and the task-specific TranslateGemma, the fine-tuned model yields gains of roughly +24% and +7% in ChrF, and +63% and +29% in BLEU, respectively.

Improvements are consistent across all language pairs. Awadhi shows strong gains in both metrics, with ChrF increasing from 35.20 to 45.60 (+10.40) and BLEU from 14.70 to 22.69 (+7.99). Bhojpuri — one of the more challenging languages in the baseline setting — exhibits moderate but consistent improvements, with ChrF rising from 35.82 to 37.60 (+1.78) and BLEU from 15.08 to 17.29 (+2.21). Chhattisgarhi and Magahi demonstrate substantial performance increases: Chhattisgarhi improves from 37.23 to 44.58 ChrF (+7.35) and from 15.36 to 21.19 BLEU (+5.83), while Magahi reaches 45.50 ChrF (+8.49 from 37.01) and 22.90 BLEU (+6.60 from 16.30), approaching the performance range of task-specific systems. Maithili also benefits from fine-tuning, although the gains are comparatively smaller than for other languages, with ChrF improving from 34.29 to 40.56 (+6.27) and BLEU from 12.38 to 16.21 (+3.83), suggesting relatively lower responsiveness to fine-tuning for this language pair.

Despite these improvements, a performance gap remains relative to the strongest task-specific systems. Encoder-decoder NMT models such as NLLB-200 and IndicTrans2 continue to achieve higher scores for several languages, particularly Chhattisgarhi and Maithili. Nevertheless, the fine-tuned Gemma-4 model substantially narrows this gap compared to its zero-shot baseline, underscoring the effectiveness of data-efficient fine-tuning for low-resource translation.

Overall, the results demonstrate that fine-tuning significantly enhances translation quality across all evaluated languages, enabling a general-purpose SLM to become a competitive translation system. The observed gains are supported by paired bootstrap resampling and are statistically significant at the $p < 0.05$ level.

Figures 2 and 3 illustrate per-language performance gains from fine-tuning.

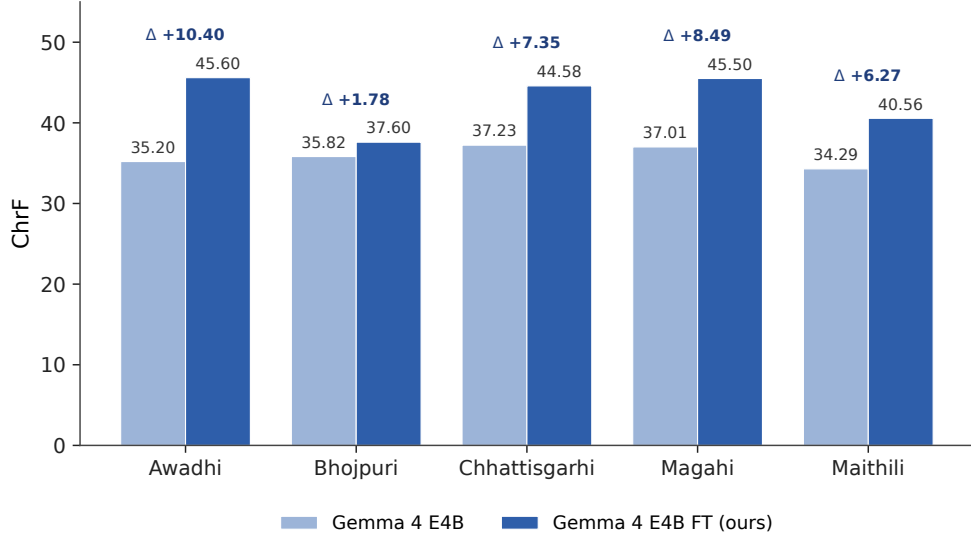


Figure 2: Per-language ChrF comparison of Gemma 4 E4B and its fine-tuned variant (ours) on FLORES-200 (EN $\rightarrow$ xx). Δ denotes absolute gains from fine-tuning.

5.3 Ablation Study

Impact of LoRA Hyperparameters

We evaluate four LoRA configurations under a multilingual fine-tuning setting, varying the adapter rank (r) and scaling factor (α). The resulting ChrF and BLEU scores are reported in Table 7.

Overall, the configuration $(r, \alpha) = (16, 32)$ achieves the highest average ChrF score (42.77), indicating that higher-rank adapters combined with stronger scaling tend to improve multilingual adaptation when trained and evaluated on the same dataset across all configurations. This configuration likely benefits from both increased representational capacity and a stronger adaptation signal during fine-tuning. Its gains are most pronounced for Awadhi and Chhattisgarhi, and it remains competitive for Bhojpuri, suggesting that these languages benefit from higher-capacity adapters in the shared multilingual setting.

At the language level, the optimal configuration varies. Magahi performs best with the more compact setting (8, 8), whereas Maithili achieves its highest ChrF with (8, 16). These variations suggest that increasing both rank and scaling may introduce language-specific trade-offs, potentially due to mild overfitting, negative transfer, or differences in alignment with the multilingual training distribution.

Based on average ChrF, we select $(r, \alpha) = (16, 32)$ as the preferred configuration. However, the observed variability across languages highlights the importance of considering language-specific behavior when applying LoRA in low-resource multilingual settings.

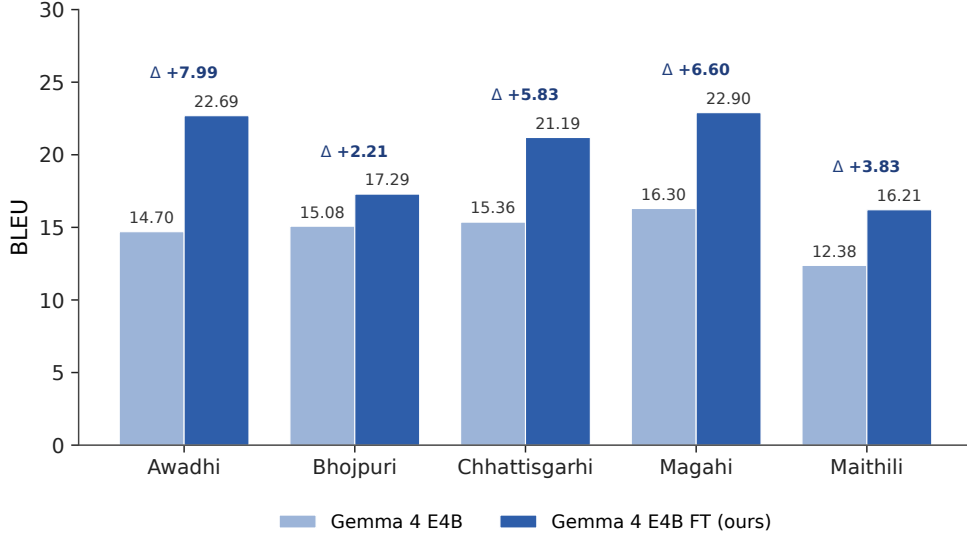


Figure 3: Per-language BLEU comparison of Gemma 4 E4B and its fine-tuned variant (ours) on FLORES-200 (EN→xx). Δ denotes absolute gains from fine-tuning.

Table 7: Ablation study of LoRA configurations across languages using ChrF and BLEU metrics.

Language	ChrF				BLEU			
	($r=8$, $\alpha=8$)	($r=8$, $\alpha=16$)	($r=16$, $\alpha=16$)	($r=16$, $\alpha=32$)	($r=8$, $\alpha=8$)	($r=8$, $\alpha=16$)	($r=16$, $\alpha=16$)	($r=16$, $\alpha=32$)
Awadhi	40.60	40.33	38.64	45.60	18.48	18.46	16.81	22.69
Bhojpuri	37.51	37.13	37.68	37.60	16.98	16.90	17.31	17.29
Chhattisgarhi	43.18	44.15	43.13	44.58	20.31	20.85	19.91	21.20
Magahi	45.70	44.93	44.33	45.50	24.20	23.55	23.04	22.90
Maithili	41.90	43.00	42.90	40.56	19.85	20.94	20.46	16.21
Average	41.78	41.91	41.34	42.77	19.96	20.14	19.51	20.06

Monolingual vs Multilingual Fine-tuning

We compare monolingual and multilingual fine-tuning using a fixed LoRA configuration $(r, \alpha) = (16, 32)$. In the monolingual setting, a separate model is fine-tuned for each language using only its corresponding training data. In contrast, the multilingual setting trains a single model jointly on the combined data from all five languages. This comparison evaluates whether joint multilingual training yields better performance than language-specific adaptation. Table 8 reports the ChrF scores for both settings, where Δ denotes the absolute improvement of multilingual over monolingual fine-tuning.

Table 8 shows that multilingual fine-tuning consistently improves ChrF across all five languages, with an average gain of +4.39 points. This result highlights the

Table 8: Monolingual vs. multilingual fine-tuning using LoRA (r, α) = (16, 32). Δ denotes Multilingual – Monolingual.

Language	ChrF			BLEU		
	Multi	Mono	Δ	Multi	Mono	Δ
Awadhi	45.60	37.42	+8.18	22.69	17.03	+5.66
Bhojpuri	37.60	35.78	+1.82	17.29	16.10	+1.19
Chhattisgarhi	44.58	41.56	+3.02	21.19	19.64	+1.55
Magahi	45.50	38.69	+6.81	22.90	17.82	+5.08
Maithili	40.56	38.43	+2.13	16.21	16.50	-0.29
Average	42.77	38.38	+4.39	20.06	17.42	+2.64

effectiveness of joint adaptation for related low-resource Indic languages, where shared lexical, morphological, and syntactic patterns can facilitate cross-lingual transfer.

The largest improvements are observed for Awadhi (+8.18) and Magahi (+6.81), indicating substantial benefits from multilingual training. Chhattisgarhi (+3.02) and Bhojpuri (+1.82) also show clear gains, albeit smaller in magnitude, while Maithili exhibits a more modest improvement (+2.13).

Overall, the consistent gains across languages suggest that multilingual fine-tuning provides a more effective training strategy than separate monolingual models in this setting, particularly for enabling knowledge sharing across related low-resource languages.

Held-out Related Languages: Generalization and Retention

To assess whether fine-tuning preserves cross-lingual generalization, we evaluate the adapted models on Hindi and Marathi, both excluded from the fine-tuning data. These languages serve as held-out related test cases, as Indo-Aryan languages sharing substantial orthographic overlap with the fine-tuned languages through the Devanagari script. This experiment evaluates whether adaptation to low-resource Indic languages leads to catastrophic forgetting or retains the model’s multilingual capabilities.

Table 9 reports results for Hindi and Marathi, with the base (vanilla) model as the pre-fine-tuning reference. Since neither language is observed during adaptation, performance changes directly reflect retention after fine-tuning.

As shown in Table 9, fine-tuning results in consistent but moderate degradation across both held-out languages. For Hindi, ChrF decreases range from -2.02 to -3.03 , while BLEU drops range from -2.66 to -4.21 . Marathi exhibits a similar pattern, with ChrF reductions between -1.72 and -3.14 , and BLEU decreases between -2.08 and -3.45 . These relatively small losses indicate that the model retains a substantial portion of its pre-trained multilingual competence despite the absence of these languages during fine-tuning.

Across configurations, smaller adapters consistently preserve performance more effectively. The $(r, \alpha) = (8, 8)$ setting achieves the best retention for both Hindi (55.67

Table 9: Held-out results (EN→Hindi, EN→Marathi) for the fine-tuned Gemma 4 E4B model. Δ denotes change relative to the base model.

Setting	Hindi				Marathi			
	ChrF	Δ	BLEU	Δ	ChrF	Δ	BLEU	Δ
Base Model	57.69	–	38.17	–	51.14	–	28.14	–
<i>Fine-tuned Model Variants</i>								
LoRA ($r=8, \alpha=8$)	55.67	-2.02	35.51	-2.66	49.42	-1.72	26.06	-2.08
LoRA ($r=8, \alpha=16$)	55.39	-2.30	35.00	-3.17	49.08	-2.06	25.62	-2.52
LoRA ($r=16, \alpha=16$)	55.53	-2.16	35.29	-2.88	49.16	-1.98	25.70	-2.44
LoRA ($r=16, \alpha=32$)	54.66	-3.03	33.96	-4.21	48.00	-3.14	24.69	-3.45

ChrF, -2.02) and Marathi (49.42 ChrF, -1.72), corresponding to approximately 96% and 97% retention of the base model’s ChrF scores, respectively. In contrast, the largest configuration (16, 32) shows the greatest degradation across both languages. This trend suggests a capacity–retention trade-off: higher-capacity adapters enable stronger task adaptation but may induce greater drift from the base model’s multilingual representations.

Importantly, the degradation patterns are consistent across both Hindi and Marathi, indicating that the observed effects reflect a systematic shift in cross-lingual behavior rather than language-specific variation. The relatively modest performance gaps further suggest that fine-tuning does not lead to severe over-specialization.

Overall, these results demonstrate that the proposed fine-tuning strategy maintains substantial generalization to related unseen languages, with only limited performance loss. This provides evidence that low-rank adaptation can balance specialization for low-resource languages with retention of multilingual knowledge.

6 Linguistic Analysis

This section analyses the linguistic properties of model outputs, focusing on morphosyntactic accuracy, dialectal consistency, and cross-lingual interference. Unlike automatic metrics, this analysis reveals systematic errors and adaptation patterns that are critical for low-resource language translation.

6.1 Awadhi

The base model does not revert to Hindi, but instead collapses into other Devanagiri languages, with outputs showing mixed Bhojpuri and Maithili morphology (e.g., अछि, बा, एगो, जे, आ) often within a single sentence. This indicates the absence of a stable Awadhi representation and reliance on a pooled Eastern Hindi-like space.

Fine-tuning substantially reduces this mixing and introduces core Awadhi features. The conjunction अउर becomes dominant (replacing आ and और), and the copula system shifts from अछि to अहै/अहैं. Additional gains include use of the demonstrative इ and emergence of honorific past forms (कहिन, दिहिन, रहिन).

However, two limitations remain: occasional substitution of Hindi है for अहै, and residual use of the Bhojpuri copula बा. Table 10 illustrates the reduction of cross-dialect contamination after fine-tuning.

Table 10: English→Awadhi translation example. **Red** = base model (contaminated); **blue** = fine-tuned model. Key dialect markers are underlined.

	Text
English	<i>Once the ancient city of Smyrna, now a modern, busy commercial centre built around a large bay and surrounded by hills.</i>
Base	पुरानी स्मिर्ना शहर, जे <u>अहाँ</u> के एगो आधुनिक, विकसित, आ व्यस्त वाणिज्य केंद्र ह, जे <u>एगो</u> बड़ खातिर खाड़ी के इर्द-गिर्द बसावल बा <u>आ</u> पहाड़न से घिरे बा। (Mixed Maithili + Bhojpuri: relative जे×2; pronoun अहाँ; article एगो×2; copula बा×2; conjunction आ×2. Nine cross-dialect markers, zero Awadhi markers.)
FT	एक समय इ स्मिर्ना नाम के प्राचीन शहर रहा, आज एक आधुनिक, विकसित <u>अउर</u> व्यस्त वाणिज्यिक केंद्र बन गइस <u>अउर</u> एक बहुत बड़ खाड़ी के चारों ओर बसा <u>अउर</u> पहाड़न से घिरल बा। (Awadhi demonstrative इ; conjunction अउर×3 replacing Maithili आ; past-tense गइस; oblique plural पहाड़न. Five Awadhi markers introduced; one residual Bhojpuri copula बा persists.)

6.2 Bhojpuri

The base model largely defaults to Hindi, with key grammatical slots filled by Hindi function words (e.g., और, है, को, के लिए) and little consistent Bhojpuri morphology. Occasional Bhojpuri forms (e.g., बा, आ) appear alongside Hindi, resulting in unstable code-mixing.

Fine-tuning produces a clear morphosyntactic shift: Hindi markers are largely removed, while Bhojpuri features become dominant. The copula बा, adversative बाकिर, and conjunction आ increase, along with productive habitual-present forms in -ला/-एला (e.g., जाला, करेला, होला). This indicates acquisition of paradigmatic Bhojpuri inflection rather than lexical substitution.

However, one limitation persists: repetition-loop hallucinations, a known effect of PEFT under skewed data distributions. Table 11 illustrates the shift from Hindi to consistent Bhojpuri morphology.

6.3 Chhattisgarhi

the base model exhibits two failure modes: Hindi fallback and cross-dialect contamination (particularly Maithili forms such as अछि, सँ). These often co-occur, reflecting an unstable dialect representation.

Fine-tuning addresses both failure modes. The most notable gain is acquisition of the Chhattisgarhi pluraliser मन, used productively across nouns. The copula हवय and

Table 11: English→Bhojpuri translation example (FLORES devtest idx 592). **Red** = base model (Hindi output); **blue** = fine-tuned model. Key Bhojpuri markers are underlined.

	Text
English	<i>But experimentation alone is not enough—an experiment is an investigation used to eliminate hypotheses; questioning and observation also guide scientific research.</i>
Base	लेकिन अकेले नहीं है, प्रयोग, और एक प्रयोग एक जांच है जिसका उपयोग एक या एक से अधिक संभावित परिकल्पनाओं को खत्म करने के लिए किया जाता है, सवाल पूछना, और अवलोकन करना भी वैज्ञानिक अनुसंधान का मार्गदर्शन करते हैं। (Pure Hindi: adversative लेकिन; copula है×3; conjunction और×2; dative को; purposive के लिए. Zero Bhojpuri markers.)
FT	<u>बाकिर</u> , ई अकेल ना ह, प्रयोग बा, आ एगो प्रयोग एगो अइसन जांच ह जे संभावित परिकल्पना में से एगो चाहे ढेर बाटे के खारिज करे में इस्तेमाल करे जाला, सवाल पूछे में आउर अवलोकन करे में भी वैज्ञानिक खोज के दिशा देला. (Adversative बाकिर replaces लेकिन; copula बा replaces है; article एगो×3; conjunction आ/आउर; habitual-present जाला, देला. Ten Bhojpuri markers introduced; zero Hindi markers remain.)

conjunction अउ increase, while Hindi और is nearly eliminated. Habitual forms in -थे and the dative ला align more closely with reference usage.

However, one limitation persists: the purposive marker बर is underrepresented, while खातिर is overused, suggesting substitution from neighboring systems. Table 12 illustrates the shift from Maithili contamination to consistent Chhattisgarhi morphology.

Table 12: English→Chhattisgarhi translation example (FLORES devtest idx 675). **Red** = base model (Maithili contamination); **blue** = fine-tuned model. Key Chhattisgarhi markers are underlined.

	Text
English	<i>It has long been known that various types of brain injury, trauma, lesions, and tumors affect behaviour and alter some mental functions.</i>
Base	लाम्बे समय सँ जानल अछि जे विभिन्न प्रकारक दिमागक क्षति, आघात, घाव, आ ट्यूमर व्यवहार पर असर करैत अछि आ कुछ मानसिक कार्यमे परिवर्तन करैत अछि। (Pure Maithili: copula अछि×3; postposition सँ; conjunction आ×2; genitive suffix -क. Zero Chhattisgarhi markers.)
FT	लम्बा समय ले जानल जात हवय कि दिमाग के अलग-अलग प्रकार के नुकसान, आघात, घाव अउ ट्यूमर व्यवहार ला प्रभावित करत हवय अउ कुछ मानसिक कार्य मन म परिवर्तन करत हवय। (CG copula हवय×3; conjunction अउ×2; dative ला; ablative ले; noun pluraliser मन. Eight CG markers introduced; all Maithili markers eliminated.)

6.4 Magahi

Magahi shows the most complex behavior. The base model produces mixed Hindi, Bhojpuri, and Maithili outputs, with Hindi function words dominant and sporadic intrusion of Bhojpuri habitual forms and Maithili markers.

Fine-tuning removes most Hindi and Bhojpuri layers while increasing Magahi-specific features such as the copula हय, conjunction अउ, purposive लेल, and negative नय, indicating acquisition of productive morphology.

However, repetition-loop hallucinations persist.

Table 13 illustrates the shift from Hindi to mixed Magahi output.

Table 13: English→Magahi translation example (FLORES devtest idx 955). **Red** = base model (Hindi output); **blue** = fine-tuned model. Key Magahi markers are underlined.

	Text
English	<i>The official languages of Barcelona are Catalan and Spanish. About half the people prefer speaking Catalan, a vast majority understands it, and virtually everyone knows Spanish.</i>
Base	बार्सिलोना के आधिकारिक भाषा कैटलन और स्पैनिश है। लगभग आधा लोग कैटलन बोलना पसंद करते हैं, एक बड़ा बहुमत इसे समझता है, और लगभग हर कोई स्पैनिश जानता है। (Pure Hindi: conjunction और×2; copula है×3; habitual -ता है×2. Zero Magahi markers.)
FT	बार्सिलोना के आधिकारिक भाषा कैटलन <u>अउ</u> स्पेनिश <u>हय</u> । लगभग आधा लोग कैटलन बोलब पसंद करैत <u>हय</u> , एक बहुत बड़ संख्या <u>ओकरा</u> समझल <u>हय</u> , <u>अउ</u> लगभग सब लोग स्पेनिश जानइ <u>हय</u> । (Magahi copula हय×4; conjunction अउ×2; oblique pronoun ओकरा. Seven Magahi markers introduced from a fully Hindi base; zero Maithili interference.)

6.5 Maithili

Maithili exhibits the most severe base-model failure, with outputs often in entirely different languages (notably Nepali and Marathi), alongside some Hindi-mixed outputs. This reflects a breakdown in language identity rather than dialect selection.

Fine-tuning eliminates cross-language outputs and restores consistent Maithili usage. Core markers such as the copula अछि, article एकटा, conjunction आ, locative मे, ablative सँ, and negative नहि increase substantially. Honorific verb forms (-नि/-थि) are also acquired, indicating learning of register-sensitive morphology.

However, two limitations remain: overuse of सँ and honorific forms, and a higher frequency of repetition-loop hallucinations compared to other languages. Table 14 illustrates the correction from incorrect-language outputs to consistent Maithili.

6.6 Cross-language observations

Across all five languages, several consistent patterns emerge:

Table 14: English→Maithili translation example. **Red** = base model (Marathi output); **blue** = fine-tuned model. Key Maithili markers are underlined.

	Text
English	<i>Dr. Ehud Ur, a professor of medicine at Dalhousie University in Halifax ... cautioned that the research is still in its early days.</i>
Base	डाक्टर एहुद अर, हैलिफैक्स, नोवा स्कोटिया मधील डलहाउस-ई विद्यापीठाचे वैद्यकीय प्राध्यापक आणि कॅनेडियन डायबेटिस असोसिएशनच्या क्लिनिकल आणि सायंटिफिक विभागाचे अध्यक्ष यांनी सावध केले की हे संशोधन अजून सुरुवातीच्या टप्प्यात आहे. (Full Marathi: <i>genitive -चे/-च्या; conjunction आणि×2; copula आहे; ergative यांनी. Zero Maithili morphology.</i>)
FT	हलिफाक्स, नोवा स्कोटिया में दलहौसी विश्वविद्यालय में चिकित्साक प्राध्यापक डा. एहुद उर आ कॅनेडियन डायबेटिस एसोसिएसनक क्लिनिकल आ साइंटिफिक डिविजनक अध्यक्ष चेतावनी देलन कि अनुसन्धान अछि एहि के सुरु कयल गेल दिन सभ में। (<i>Maithili conjunction आ×2; copula अछि; genitive suffix -क×3; locative में×2. Eight Maithili markers from a fully Marathi base.</i>)

- Outputs are strongly biased by pretraining toward higher-resource or related languages, often leading to partial or full substitution.
- Fine-tuning drives coordinated shifts at the morphological level, affecting copulas, conjunctions, postpositions, and inflection rather than isolated words.
- Cross-dialect interference persists, with models substituting markers from related varieties instead of maintaining strict separation.
- Frequent dialectal markers tend to be overgenerated after fine-tuning, often accompanied by repetition or hallucination in low-resource conditions.

These findings highlight key bottlenecks: improving dialect-specific data quality and expanding training coverage beyond limited datasets.

7 Limitations

Our evaluation relies on lexical overlap metrics (BLEU and ChrF), which may systematically underestimate translation quality when outputs are semantically correct but lexically divergent from the reference. This is particularly relevant for closely related languages and dialects, where multiple valid phrasings or synonymous expressions exist with limited surface overlap. Learned metrics such as COMET are designed to capture such semantic equivalence; however, most publicly available COMET models rely on multilingual encoder backbones (e.g., XLM-R) trained predominantly on high-resource languages, resulting in limited representation coverage for Awadhi, Bhojpur, Chhattisgarhi, and Magahi. In this setting, COMET scores may reflect encoder biases rather than true translation quality, undermining reliable evaluation. We therefore restrict evaluation to BLEU and ChrF despite their known limitations.

The experimental setup operates in an extreme low-resource regime, utilizing only 997 sentence pairs per language from the FLORES-200 development set. While this setting effectively demonstrates the sample efficiency of LoRA, it inherently constrains

both the size and diversity of the training distribution. Consequently, the resulting models are likely to exhibit domain-specific behavior, with their generalization to out-of-domain text remaining unverified. Furthermore, the limited fine-tuning data may hinder the model’s ability to capture language-specific nuances, particularly in a multilingual setting. Increasing the volume and diversity of fine-tuning data would likely improve representation learning, robustness, and cross-domain generalization.

8 Conclusion and Future Work

This paper presents a systematic study of English→Indic translation for five under-represented Devanagari-script languages—Awadhi, Bhojpuri, Chhattisgarhi, Magahi, and Maithili—using small open-source language models (< 10B parameters). We benchmark seven SLMs against dedicated NMT systems and adapt Gemma 4 (E4B) using LoRA with fewer than 1,000 sentence pairs per language, thereby simulating an extreme low-resource setting.

Our study yields three key findings. First, task-specific training remains the dominant factor in zero-shot performance: NLLB-200 consistently outperforms general-purpose SLMs despite its smaller size. Second, parameter-efficient fine-tuning produces substantial gains: the adapted Gemma 4 significantly improves over its baseline and narrows the gap to specialised NMT systems while requiring minimal data and compute. Third, multilingual fine-tuning consistently surpasses monolingual adaptation, indicating that cross-lingual transfer across related Devanagari-script languages remains effective even under severe data constraints.

Beyond improvements in automatic metrics, qualitative analysis reveals that fine-tuning enhances translation at the morphological level, enabling the recovery of language-specific copulas, agreement patterns, and postpositions rather than relying primarily on surface-level lexical substitutions. Evaluation on held-out related languages further suggests that such adaptation preserves a significant degree of multilingual generalisation.

Overall, these results demonstrate that compact SLMs, when combined with parameter-efficient adaptation techniques, offer a practical and scalable pathway for developing machine translation systems for underrepresented Indic languages.

This work also highlights several promising directions for future research. First, extending evaluation to additional low-resource Indic languages (e.g., Bundeli, Braj, Dogri, and Kumaoni), as well as to other script families, would help assess the generality of our findings. Second, benchmarking newer model families may provide further insight into trade-offs in multilingual performance and efficiency.

We hope this work encourages the use of data-efficient adaptation strategies to extend language technologies to the long tail of underrepresented languages.

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